Assignment 2: MNIST Classification using PCA, FDA, and Discriminant Analysis

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Problem Statement

The objective of this assignment is to build a classification pipeline for MNIST digits 0, 1, and 2, utilizing dimensionality reduction techniques before classification. The task involves:

- Randomly sampling 100 training and 100 testing images per class (I have used seed 0 for sampling so that the accuracy does not vary every time. We can however do it randomly by removing the seed).
- Implementing Principal Component Analysis (PCA) to reduce dimensionality while retaining specific variance thresholds (75% and 90%) and computing reconstruction Mean Squared Error (MSE).
- Implementing Fisher's Discriminant Analysis (FDA) finding direction that maximizes class separation.
- Using Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) to classify the reduced test samples.
- Visualizing and comparing the 2D transformed feature spaces for both PCA and FDA.

Mathematical Equations Used

Principal Component Analysis (PCA)

PCA maximises the variance of the projected data. For a data matrix $X \in \mathbb{R}^{d \times N}$, we first center the data:

$$X_c = X - \mu$$

The covariance matrix S is computed as:

$$S = \frac{1}{N-1} X_c X_c^T$$

We solve the eigenvalue problem $Su = \lambda u$ to find the principal components. The projection to a lower-dimensional space $\mathbb{R}^p$ is:

$$Y = U_p^T X_c$$

p is found according to the variance(v) as:

$$\frac{\sum_{i=1}^{p} \lambda_i}{\sum_{i=1}^{d} \lambda_i} \geq v \tag{1}$$

Where U_p contains the top p eigenvectors. The reconstructed data $\hat{X}$ and Mean Squared Error (MSE) of a particular observation $x_{d\times 1}$ with its corresponding reconstructed observation $\hat{x}_{d\times 1}$ is:

$$MSE = \frac{1}{d}\sum_{j=1}^{d}(x_j - \hat{x}_j)^2 \tag{2}$$

Fisher's Discriminant Analysis (FDA)

FDA aims to maximize the between class variance and minimises the within class variance.

Within-class scatter matrix (S_W):

$$S_W = \sum_{c=1}^{C} \sum_{x \in \text{class } c} (x - \mu_c)(x - \mu_c)^T$$

Between-class scatter matrix (S_B):

$$S_B = \sum_{c=1}^{C} N_c(\mu_c - \mu)(\mu_c - \mu)^T$$

The projection matrix W ($W \in \mathbb{R}^{d\times 2}$) is found by computing the eigenvectors corresponding to the largest eigenvalues of $S_W^{-1}S_B$. The projected data is $Y = W^T X$.

Implementation Details

- **Data Selection:** Digits 0, 1, and 2 were filtered. 100 samples per class were randomly selected for both training and testing (Total Train: 300, Total Test: 300).
- **Preprocessing:** Images were flattened from 28×28 matrices to 784-dimensional vectors and normalized to the range $[0, 1]$ by dividing by 255.0.
- **Numerical Stability:** A small regularization term ($= 0.0001$) was added to the diagonal of covariance matrices and S_W prior to inversion to prevent singular matrix errors.
- **Classification:** LDA and QDA models were trained on the reduced dimensions Y obtained from both PCA and FDA.

Classification Accuracy

The table below displays the testing accuracies for LDA and QDA across different dimensionality reduction strategies.

Linear Discriminant Analysis (LDA) Results

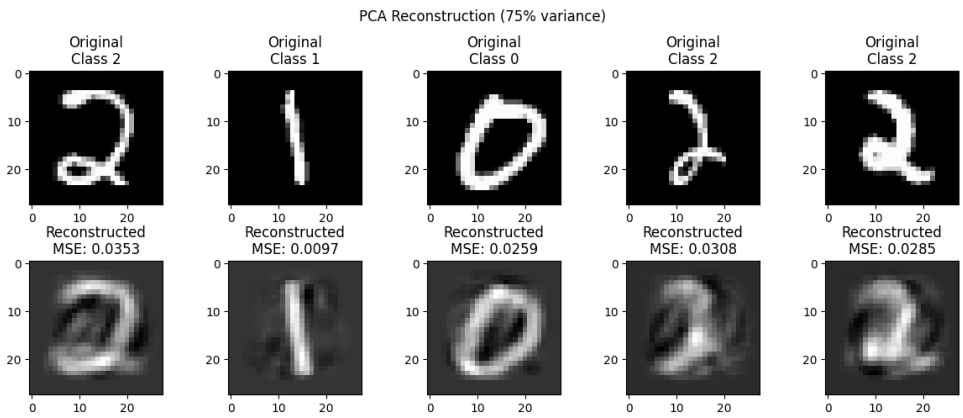
Dimensionality Reduction Method	Train Accuracy	Test Accuracy
PCA (75% Variance Retained)	97.34%	97.67%
PCA (90% Variance Retained)	99%	96.67%
PCA (2 Components)	90.67%	93.33%
FDA	95.34%	95.67%

Quadratic Discriminant Analysis (QDA) Results

Dimensionality Reduction Method	Train Accuracy	Test Accuracy
PCA (75% Variance Retained)	100%	96%
PCA (90% Variance Retained)	100%	94%
PCA (2 Components)	95%	92.67%
FDA	96.67%	95.67%

PCA Image Reconstruction

Using the PCA retaining 75% variance, 5 test samples were reconstructed back into 784-dimensional space. The Mean Squared Error (MSE) for these samples are also present in the image.



Discriminant Values for Sample Test Point

When using lda qda after using PCA with 75% variance retention. For the sample data point in the test set (Sample index: 0) the computed discriminant values are:

LDA Values: True class: 2

Class (c)	**Discriminant Score ($g_c(x)$)**
0	-8.6917
1	-11.3542
2	0.3403

The model picks the one with the highest score (Class 2).

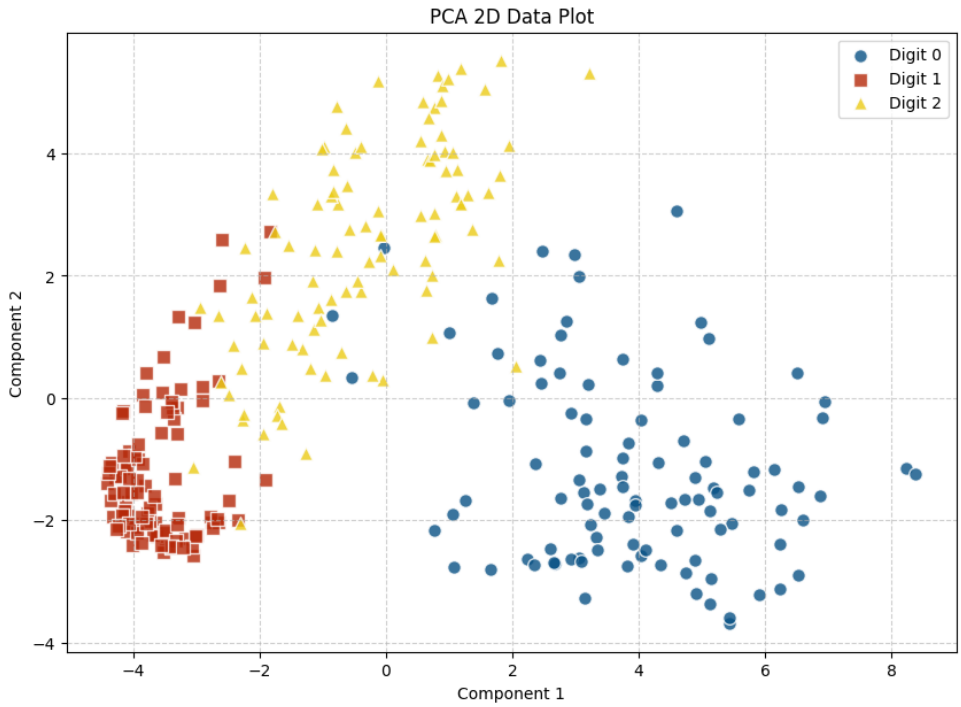
QDA Values: True class: 2

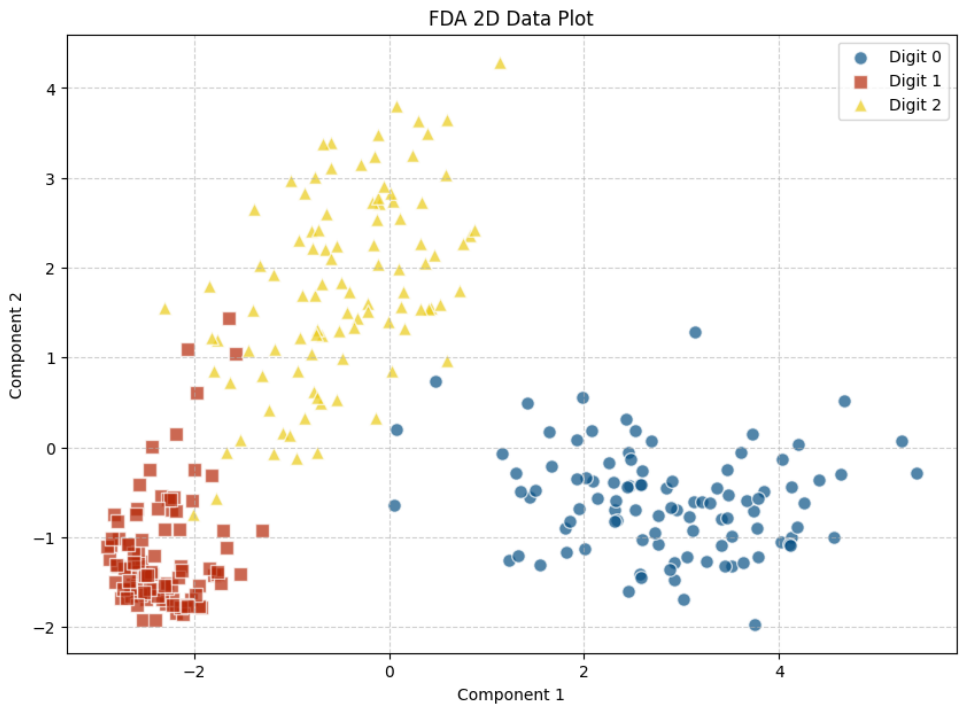
Class (c)	**Discriminant Score ($g_c(x)$)**
0	-39.4737
1	-1222.2681
2	-11.4899

The model picks the one with the highest score (Class 2).

2D Feature Space Visualization

The scatter plots below illustrate the separation of the three classes when projected onto a 2D feature space using PCA (top 2 eigenvectors) versus FDA (2 Fisher discriminants).





Analysis of PCA on Classification Performance

Principal Component Analysis (PCA) was applied before classification using LDA and QDA to see how dimensionality reduction affects classification performance. Different variance retention was done.

- **PCA with 75% Variance Retained:** Retaining 75% of the total variance provided the best accuracy as it would we removing the directions with noise. The reduced feature space captured the most important structure of the digit images while filtering out unnecessary variations. As a result, this setting produced the highest test accuracies, achieving 97.67% for LDA and 96% for QDA.
- **PCA with 90% Variance Retained:** When the retained variance was increased to 90%, the classifiers achieved higher training accuracy (99% for LDA and 100% for QDA). The test accuracy decreased slightly (96.67% for LDA and 94% for QDA). This indicates overfitting, as it might be capturing more noise and taking into account the directions which are not needed.
- **PCA with Only 2 Components:** Reducing the dimensionality to exactly two principal components significantly decreases classification accuracy (93.33% for LDA and 92.67% for QDA). Because PCA is an unsupervised method, the components with the highest variance do not necessarily correspond to the features that best separate the digit classes. The discriminative information is also lost.

Comparison Between PCA and FDA when taking only 2 components When both methods were restricted to a two-dimensional feature space, Fisher's Discriminant Analysis (FDA) achieved significantly better separation between classes than PCA. While PCA focuses on maximizing overall data variance without considering class labels, FDA explicitly maximizes the distance between class means while minimizing the variance within each class. As a result, the FDA projection produced more clearly separated clusters, leading to higher classification accuracy (95.67% for both LDA and QDA) compared to the 2D PCA representation. As we can also see in the plots that for PCA we have overlapping points but in FDA those are pushed aside as we minimise the within class scatter.